

Assignment 1.5

Day 5: Assemble and Disassemble a PC Practical (1 hour): Full hands-on session: Assemble and disassemble a desktop PC Task: Document the process with labeled steps. Pending

Task – 1

List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.



After opening the PC cabinet, the following main components can be found inside.

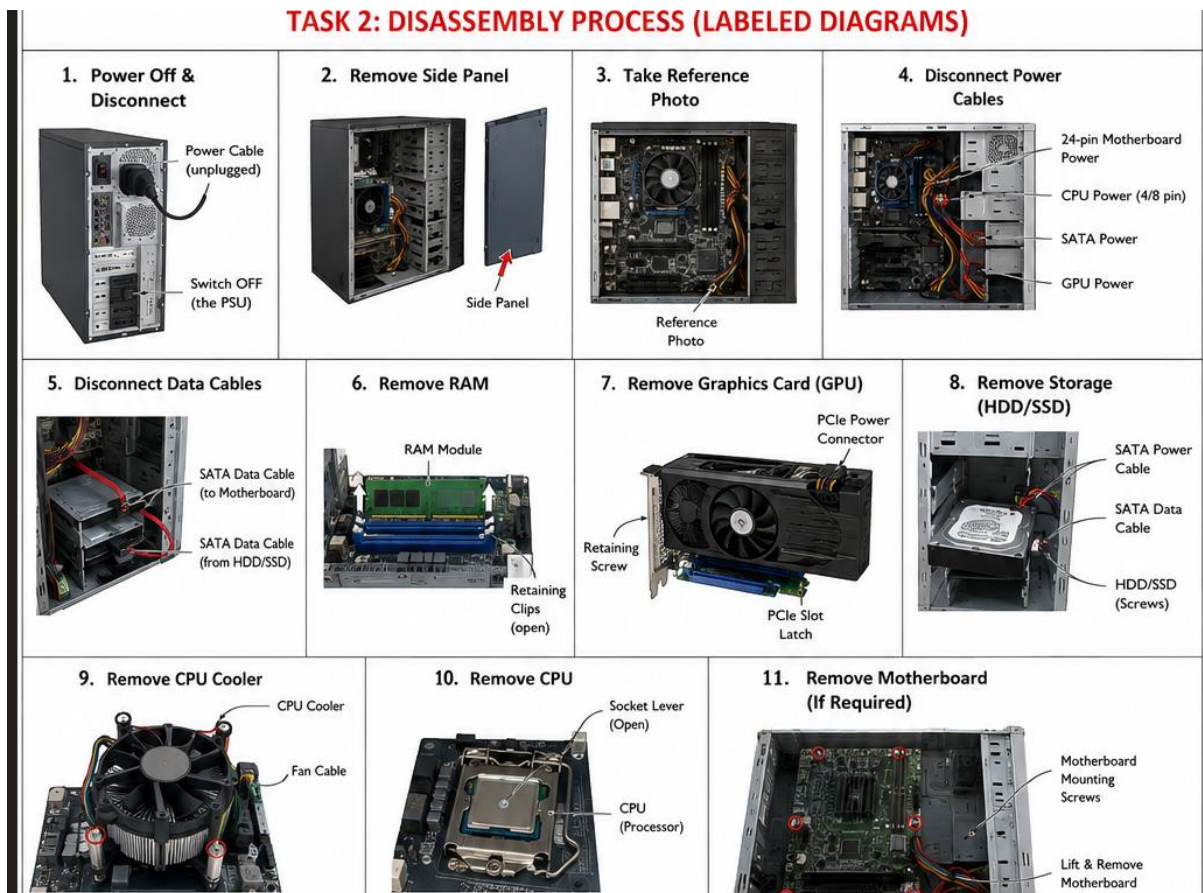
No.	Component	Short Description
1	Motherboard	The main circuit board of the computer. It connects the CPU, RAM, storage, expansion cards, and other components.
2	CPU / Processor	The main processing unit that executes instructions and performs calculations for the computer.
3	CPU Cooler / Fan	Removes heat generated by the processor and keeps the CPU at a safe operating temperature.
4	RAM	Temporary memory used by the computer to store data and programs that are currently being used.
5	Power Supply Unit (PSU)	Converts AC electricity from the wall outlet into the DC power required by computer components.
6	HDD / SSD	Provides permanent storage for the operating system, applications, and user files.
7	Graphics Card (GPU)	Processes graphics and sends video output to the monitor. Some computers use integrated graphics instead.
8	Cabinet / Case	The enclosure that holds and protects the internal computer components.
9	Case Fans	Circulate air through the cabinet to help keep internal components cool.
10	CMOS Battery	A small battery on the motherboard that maintains BIOS/UEFI settings and system time when the PC is powered off.
11	SATA Data Cable	Connects SATA storage devices such as HDDs and SATA SSDs to the motherboard.
12	Power Cables	Supply electrical power from the PSU to components such as the motherboard, drives, and graphics card.

Task – 2

Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.
Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.



The following labeled diagrams illustrate the step-by-step process of safely disassembling a desktop computer



Step 1 – Power Off and Disconnect the PC: The computer was completely shut down and disconnected from the power source before starting the disassembly process.

Step 2 – Remove Side Panel: The side panel was removed to access the internal components of the desktop computer.

Step 3 – Internal Reference Photo: A photograph of the original cable and component arrangement was taken to ensure correct reassembly.

Step 4 – Disconnect Power Cables: Power cables were carefully disconnected from the components to prepare the PC for component removal.

Common cables include:

- 24-pin motherboard power
- 4/8-pin CPU power
- SATA power
- GPU power
- Fan power

Step 5 – Disconnect SATA Data Cable: The SATA data cables connecting the storage device to the motherboard were disconnected carefully.

Step 6 – Remove RAM: The RAM retaining clips were opened and the memory module was carefully removed by holding its edges.

Step 7 – Remove Graphics Card: The graphics card was disconnected from its power supply, released from the PCIe slot, and carefully removed.

Step 8 – Remove HDD/SSD: The storage device was disconnected and removed from its mounting position inside the cabinet.

Step 9 – Remove CPU Cooler: The CPU cooling assembly was carefully removed to expose the processor.

Step 10: Remove CPU

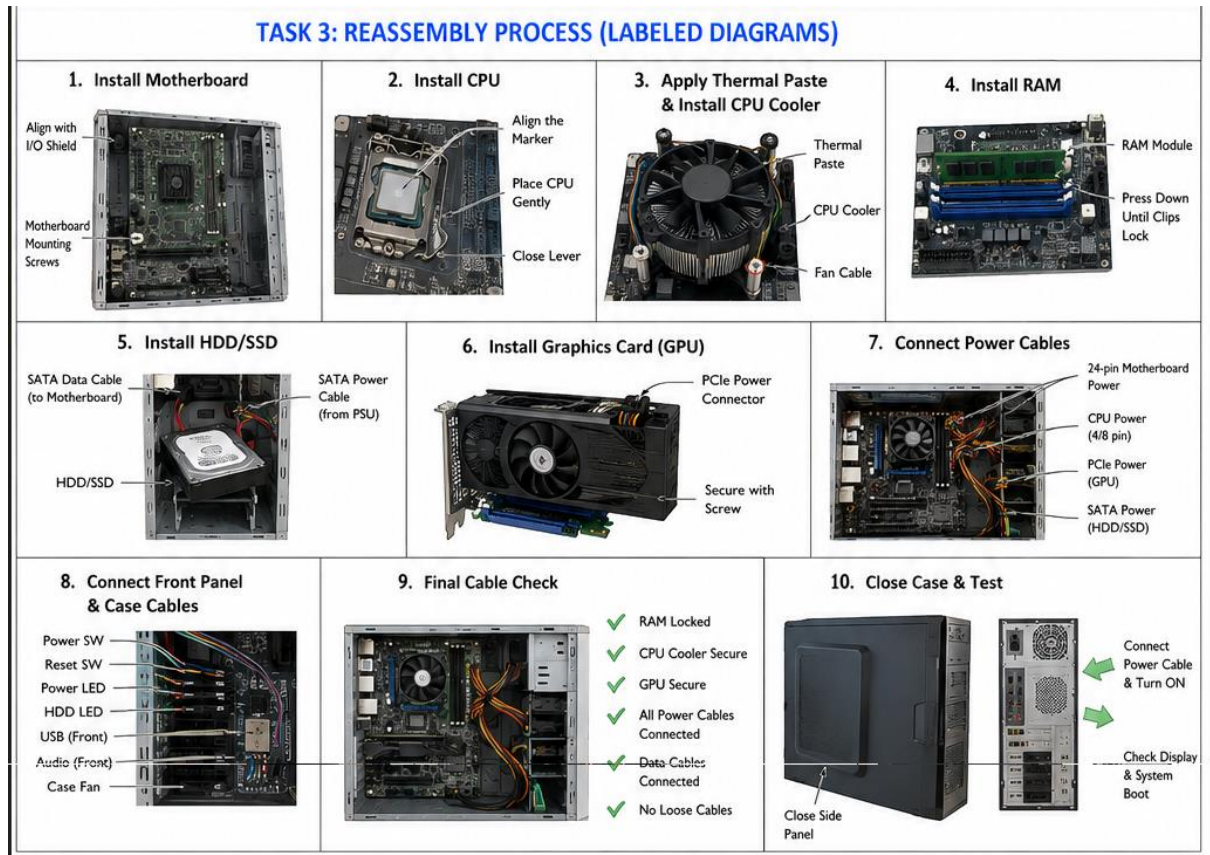
If instructed to remove the CPU:

1. Open the CPU socket retention mechanism.
2. Carefully lift the processor.
3. Hold it only by its edges.
4. Place it in a safe location.

Step 11 – Remove Motherboard: After disconnecting all cables and removing the mounting screws, the motherboard was carefully removed from the cabinet.

Task – 3

Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.



Task – 4

Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.



No.	Safety Precaution	Why It Is Important
1	Switch off the PC	Prevents electrical hazards and accidental component damage.
2	Unplug the power cable	Disconnects the PC from the electrical supply.
3	Use an anti-static wrist strap	Helps prevent electrostatic discharge from damaging sensitive components.
4	Work on a clean, dry surface	Prevents dust, moisture, and unwanted objects from damaging components.
5	Hold components by their edges	Prevents fingerprints and accidental contact with sensitive contacts.
6	Do not force connectors	Connectors should normally fit in the correct orientation; forcing them can cause damage.
7	Keep screws organized	Prevents losing screws and installing an incorrect screw in the wrong location.
8	Do not touch electrical contacts	Oils and dirt from fingers can contaminate contacts.
9	Be careful with the CPU	CPU pins/contacts can be easily damaged.
10	Do not work on a powered PC	Reduces risk of electrical shock and component damage.
11	Keep liquids away	Liquid can cause short circuits and permanent damage.
12	Check cables before powering on	Prevents incorrect connections and possible component damage.

Tool	Purpose
Phillips screwdriver	Used to remove and install most PC screws.
Anti-static wrist strap	Protects components from electrostatic discharge.
Small container/screw tray	Keeps screws organized during disassembly.
Flashlight	Helps inspect components and connectors inside the cabinet.
Compressed air / air blower	Can be used to remove dust, following appropriate precautions.
Thermal paste	Used when reinstalling a CPU cooler if the existing paste needs replacement.

Tool	Purpose
Cable ties	Used to organize cables during reassembly.
Clean workspace	Provides a safe area for handling components.